

nRF54L15 IC

Peripheral	nRF54L15 Pin	Module Pin
UIC	P1.00/XL1	25
P1.01/XL2	26	
P1.02/NFC1	27	
P1.03/NFC2	28	
P1.04/AIN0	29	
P1.05/AIN1	30	
P1.06/AIN2	31	
P1.07/AIN3	32	
P1.08/CLK16M	33	
P1.09	34	
P1.10	35	
P1.11/AIN4	36	
P1.12/AIN5	37	
P1.13/AIN6	38	
P1.14/AIN7	39	
P1.15	40	
P1.16	41	

nRF54L15 Peripherals

Accelerometer

Accelerometer Power Switch

RTD

Debug Connector

Programming Connector

Door reed block not populated on the TANK variant

Temperature Sensor

Battery Connectors

External Watchdog

Title		
Size	Number	Revision
A2		
Date:	8/1/2026	Sheet of
File:	C:\Users\Yank and Door Module Sch	Drawn by:

TANK / CARGO-TEMPERATURE Sub-node - Schematic & Pin Map (all blocks except the door reed)

nRF54L15 (QFN52) | derived from Tank_and_Door_Schematic | SmartWagon Rev.1

Pin / net map

Function	Net	nRF54L15 pin	Dir	Firmware
I2C SDA (BMA400 + STS4X)	SDA	P1.04 / AIN0	bidir	i2c21
I2C SCL (BMA400 + STS4X)	SCL	P1.05 / AIN1	bidir	i2c21
Sensor-rail enable (TPS22917 ON)	SW_AXI	P1.06 / AIN2	OUT	srail.c
BMA400 interrupt (INT1)	BMA_INT	P0.04	IN	reserved
Debug UART TX	D_TX	P1.02 / NFC1	OUT	uart20
Debug UART RX	D_RX	P1.03 / NFC2	IN	uart20
Watchdog WAKE (TPL5010->MCU)	uC_Wakeup	P0.00	IN	watchdog.c
Watchdog DONE (MCU->TPL5010)	uC_Alive	P0.01	OUT	watchdog.c
Reset (TPL5010 RSTn, R17 pull-up)	NRESET	RESET	IN	-
LF crystal 32.768 kHz	XL1 / XL2	P1.00 / P1.01	-	LFXO
RTD analog (Temp_Sense)	Temp_Sense	P1.07 / AIN3	IN	rtd.c (SAADC ch3)

Key components

Ref / Part	Bus / addr	Role
IC1 BMA400	I2C 0x14	Accelerometer - tamper / impact detect
U5 STS4X	I2C 0x4A*	Temperature - sub-node internal temp
U2 LMV321WG-7	Temp_Sense	RTD bridge amplifier -> SAADC AIN3
TL431A + 2.495V	-	Precision reference exciting the RTD bridge
U3 TPS22917	SW_AXI	Load switch - gates the VCC_AXI sensor rail
U4 TPL5010	R21=169k	External nano-power watchdog
BATT1 ER34615	-	3.6 V primary Li-SOCl2 cell

BLE payload mapping (struct sw_adv_pt)

Field	Meaning
value	RTD cargo temperature x10 (degC) [primary]
value2	STS4X internal temperature x10 (degC)
flags	ALARM (over-temp) IMPACT LOWBATT

Assumptions to confirm on the board

- STS4X I2C address 0x4A assumed - confirm the exact STS4x variant.
- RTD element type (PT100 vs PT1000) and op-amp gain NOT determinable from the drawing -
 - set the two-point calibration RTD_CAL_M / RTD_CAL_B in app_config.h after a bench cal.
- SAADC channel uses AIN3 = P1.07; confirm the NRF_SAADC_AINx binding token for your nRF Connect SDK version.
- ADC gain/reference in app.overlay are a starting point - tune with the calibration above.
- TPL5010 interval set by R21=169k - firmware feeds DONE every report cycle.